

Vulnerability Assessment Report

Cyber Security Task 1 – Vulnerability Assessment

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Executive Summary

A vulnerability assessment was conducted on the Damn Vulnerable Web Application (DVWA) running in a controlled local environment. The objective was to identify common web application security flaws and document their impact and remediation methods.

Vulnerability 1: SQL Injection

Vulnerability Name

SQL Injection (SQLi)

Risk Level

High

Description

SQL Injection occurs when user-supplied input is directly incorporated into SQL queries without proper validation or parameterization. This allows attackers to manipulate database queries and access unauthorized information.

Test Location

DVWA → SQL Injection Module

Payload Used

1' OR '1'='1

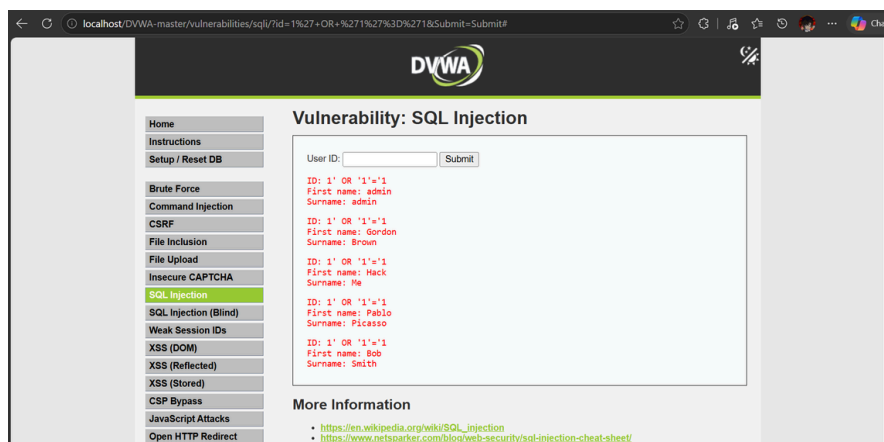
Proof of Concept

The payload was entered into the User ID field. Instead of returning a single user record, the application returned multiple records from the database, confirming successful SQL

injection.

Evidence

[Insert Screenshot of SQL Injection Result Here]



Impact

- Unauthorized access to sensitive information
- Authentication bypass
- Database disclosure
- Potential modification or deletion of records

OWASP Category

OWASP Top 10 2021 – A03: Injection

Recommended Mitigation

- Use Prepared Statements
- Implement Parameterized Queries
- Validate User Input
- Apply Principle of Least Privilege
- Use Web Application Firewalls (WAF)

Conclusion

The assessment successfully identified a SQL Injection vulnerability within the test environment. The vulnerability demonstrates how improper input validation can lead to unauthorized database access and sensitive information disclosure. Organizations should implement secure coding practices and regular security testing to mitigate such risks.

References

- OWASP SQL Injection Guide

- DVWA Documentation
- OWASP Top 10 (2021)

Vulnerability 2 :Reflected Cross-Site Scripting (XSS)

1. Introduction

This report documents the identification and testing of a Reflected Cross-Site Scripting (XSS) vulnerability in the Damn Vulnerable Web Application (DVWA). The assessment was conducted in a controlled local environment for educational and security awareness purposes.

2. Vulnerability Overview

Vulnerability Name

Reflected Cross-Site Scripting (XSS)

Severity Level

Medium to High

OWASP Category

A03:2021 – Injection

Description

Reflected Cross-Site Scripting (XSS) occurs when an application includes untrusted user input in a web page without proper validation or output encoding. Attackers can inject malicious JavaScript code that executes in the victim's browser.

3. Testing Environment

Application

Damn Vulnerable Web Application (DVWA)

Web Server

Apache (XAMPP)

Database

MySQL

Operating System

Windows 11

Security Level

Low

4. Vulnerability Testing

Module Tested

XSS (Reflected)

Payload Used

Objective

To determine whether the application properly validates and sanitizes user input before displaying it on the webpage.

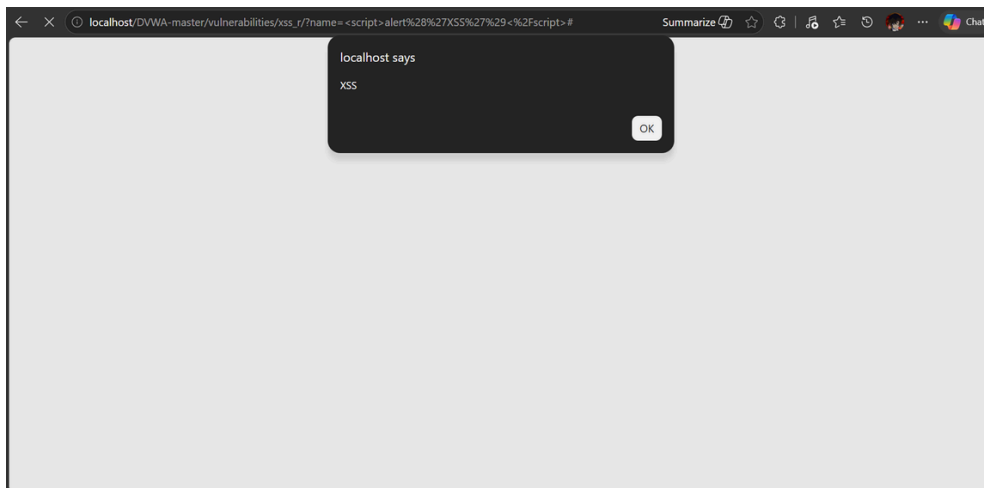
5. Proof of Concept

The payload was entered into the input field of the Reflected XSS module. After submission, the application executed the injected JavaScript code and displayed a browser alert box containing the text "XSS".

This confirms that arbitrary JavaScript can be executed within the browser context.

Evidence

[Insert Screenshot Showing the XSS Popup Alert]



6. Impact Assessment

Successful exploitation of Reflected XSS may allow attackers to:

- Execute malicious JavaScript code
- Steal session cookies
- Hijack authenticated sessions
- Redirect users to malicious websites
- Conduct phishing attacks
- Modify webpage content

7. Risk Rating

Severity: Medium-High

Likelihood: High

Business Impact: Medium-High

Overall Risk: High

8. Recommended Mitigation

1. Validate all user inputs.
2. Implement output encoding.
3. Apply Content Security Policy (CSP).
4. Sanitize user-generated content.
5. Use secure frameworks with built-in XSS protection.
6. Perform regular security testing and code reviews.

9. Conclusion

The Reflected XSS vulnerability was successfully identified and exploited in the DVWA environment. The application's failure to sanitize user input allowed arbitrary JavaScript execution within the browser. Proper input validation, output encoding, and security controls should be implemented to mitigate this risk.

References

- OWASP Cross Site Scripting (XSS) Prevention Cheat Sheet
- OWASP Top 10 (2021)
- DVWA Documentation